A-A-2539
MAY 16, 1988

COMMERCIAL ITEM DESCRIPTION

'HAMMER, HAND, BRICKLAYERS'

The General Services Administration has authorized the use of this commercial item description in lieu of Type III, 1, Styles A and B, Federal Specification GGG-H-86.

SALIENT CHARACTERISTICS

The hammers shall conform to ANSI B173.6 and requirements stated herein.

Finish. Polished striking face/protective coating of the entire head to deter rusting.

Handle. Fiberglass with handgrip or clear lacquered hickory.

Hardness. The face and chisel shall be hardened to not less than 50 nor more than 57 on the Rockwell C scale. The hardness test shall be conducted in accordance with ASTM E 18.

Eye number. The eye number and dimensions for Style A, maybe furnished maybe furnished as a number 1 or 3 and shall be as specified in figure 3.

Identification marking. Each hammer shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, and the country of origin.

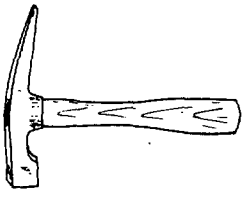


Figure 1.--Style A, Hickory.

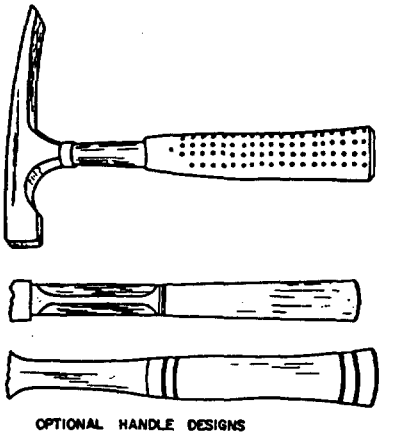


Figure 2.--Style B, Fiberglass.

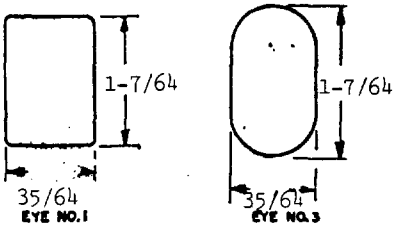


Figure 3.--Eye numbers.

(Figures are illustrative and not restrictive)

Head Weight	Handle Type	Handle Length Overall	National Stock Number
24 +-2 ozs.	Hickory	8 +-1 inch	5120-00-236-3730
24 +-2 ozs.	Fiberglass	8 +-1 inch	5120-00-900-6106

Testing:

Hammer/head alignment. The angle of the handle shall be no more than 2 degrees from the perpendicular when measured from the longitudinal centerline of the handle to a plane parallel to the top of the head.

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FSC 5120

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In addition, the rotational variance around the longitudinal centerline of the handle shall be no more than 2 degrees when viewed from the butt end of the handle along the longitudinal centerline of the handle and when measured from the longitudinal centerline of the head to a line corresponding to the long dimension of the butt end of the handle.

Crack detection test. No hammer provided under this commercial item description shall exhibit any indication of cracks in the hammer head. All samples shall be 100 percent inspected and subjected to either a wet magnetic particle inspection test in accordance with ASTM E709-80 or an Ultrascopic detection test in accordance with ASTM E114.75. All surfaces shall be examined. Any hammer head exhibiting indications of cracks shall be rejected. Hammer heads with only surface crack indications may have the indications removed by minor rework and then retested as above.

Handgrip test. Hammers with handgrips shall be subjected to this test. Each sample shall be conditioned climatically to 30 degrees F, plus or minus 2 degrees F, and remain at this temperature for 1 hour or until thermal stabilization has occurred. The sample shall then be cycled to plus 120 degrees F, plus or minus 2 degrees, and remain at this temperature for 1 hour or until thermal stabilization has occurred. The sample shall then be conditioned to plus 70 degrees F, plus or minus 10 degrees F (room ambient temperature).

The samples shall then be subjected to a pull test force applied to the grip parallel to the longitudinal axis of the handle with the hammer head securely captured to restrict movement. In order to minimize normal forces to the handle, the handgrip shall be captured using a woven wire cable grip or suitable device.

The test pull force is to be as noted below, except that for certain sizes where the handgrip pull test force is found to be greater than the pull-apart force (tensile force). In this instance the handgrips shall be subjected to the head pull-apart force (tension force) as a maximum in lieu of the specified force. Grips which loosen or separate from the handle shall be cause for rejection.

Hammer head Weight	Pull Test Force	Torque Test
Less than 16 oz.	500 lbs.	40 in-lbs.
16 oz. and over	700 lbs.	50 in-lbs.

After application of the pull test, the hammers shall be held by the head, and the grip twisted at the normal handgrip position by hand in alternating directions with a torque as noted below. Five complete twisting cycles shall be performed, after which there shall be no grip looseness or separation from the handle.

Pull apart head/handle test. All samples shall be 100 percent inspected in accordance with ANSI B173.1.

NSN 5120-00-900-6106 shall meet the testing requirements of the following overstrike test and the bending moment test:

Fiberglass handle overstrike test. Hammers with fiberglass handles shall withstand 23 full swinging overstrike blows by an average built man (160 to 180 pounds), commensurate with the end use and weight of the hammer. The blows shall be against that portion of the handle directly behind the head and on a substantially supported steel object having a minimum 3/8 inch full radius.

Fiberglass bending moment test. Hammers with fiberglass handles shall be subjected to the test below in lieu of the static force test found in ANSI B173.6. Samples of the assembled striking tool shall not break, chip, or crack when subjected to the applicable load and distance listed below. The distance shall be measured from the top of th tool head. The striking tool head shall be locked securely in the test fixture with the striking face down and with the handle extended in the horizontal plane. The load shall be applied gradually and in a vertical direction.

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BENDING MOMENT TEST LOAD FOR FIBERGLASS HANDLED HAMMERS

Head weight (ounces)	Distance (inches)	Applied load (pounds minimum)
7	10	100
13	10	200
16	10	300
20	10	325

Workmanship. All hammers shall be free from imperfections which may affect their safety, appearance, serviceability or durability.

QUALITY ASSURANCE PROVISIONS

Responsibility for inspection. Unless otherwise stated in the contract, the contractor is responsible for the performance of all inspections as specified herein. Except as stated otherwise in the contract, the contractor may use his own or any other facilities available for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform the inspections set forth in the commercial item description if deemed necessary to assure supplies and services conform to the prescribed requirements.

Sampling for inspection and testing. Sampling for inspection and testing shall be in accordance with MIL-STD-105 at the inspection levels and AQL's specified below. The lot shall consist of all hammers of the same nominal head weight presented for inspection at the same time. The sample unit shall be one hammer.

Visual and dimensional characteristics, and workmanship. The inspection level shall be S-4, with an AQL of 4.0 expressed in terms of percent defective.

Testing of the end item for hardness. The inspection level shall be S-3 with an AQL of 4.0 expressed in terms of percent defective.

Testing of the end item (all other tests). The inspection level shall be S-3 with an AQL of 4.0 expressed in terms of percent defective.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or purchase order.

Notes:

The issue of referenced documents in effect on the date of invitation for bid or request for proposals shall be used to determine compliance with the stated requirements.

ANSI standards are available from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.

ASTM standards are available from the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.

Military standards are available from the Naval Publications and Forms Center, 5801 Tabor Ave., Philadelphia, PA 19120.

MILITARY INTEREST:

Custodian

Air Force - 99
Army - GL
Navy - SH

PREPARING ACTIVITY:

GSA - FSS

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